iLearn Practice Exam

11BR · Chapter / Topic 7

Cambridge AS Level Computer Science 9618

Paper 1 Theory Fundamentals

Topic 7: Ethics and Ownership

Name: _____

Class: _____

Date: _____

Time allowed: 1 hour 10 minutes

TOTAL MARKS: 59

Marks obtained: ..... / 59

INSTRUCTIONS

· Answer all questions.
· Write your answers in the spaces provided on this paper, using a black or dark blue pen.
· Do not use an erasable pen or correction fluid.
· No calculators are allowed.
· Do not write anything in the margins.

INFORMATION

· The total mark for this paper is 59.
· The number of marks for each question or part question is shown in brackets [ ].
· This paper contains 12 questions, all drawn from Cambridge past papers on this chapter.
· After you finish, your teacher will mark your work against the official mark scheme.
· Only 59 marks of past-paper questions are available for this chapter, so the recommended time is 1 hour 10 minutes (1

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1 A software developer is working in a team writing a program for a client.

(a) One section of the program being developed will convert user's speech into commands.

Explain how Artificial Intelligence (AI) can be used in this program.

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..... [3]

2 A programmer is developing a computer game in a high-level language to sell to the public.

(a) The programmer needs to publish the game under a software licence so that it can be sold to the public.

Identify the **most appropriate** type of software licence for the game **and** justify your choice.

Licence .....

Justification .....

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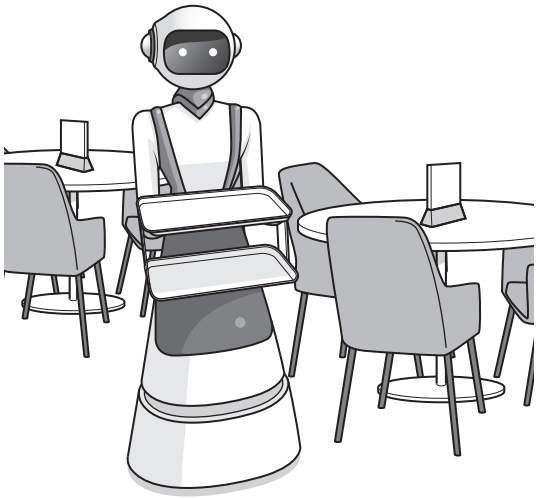
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[4]

3 Robots are used to serve food and drink to customers at a restaurant.



(a) The robot uses Artificial Intelligence (AI) to communicate with the customers. The customers speak to the robot to order their food and drinks.

Explain how AI will be used in this part of the robot.

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4 A computer program uses a digital camera to read the words on an item.

The program can read the words that are written on the item, translate the words to a chosen language and then output the words as audio. For example, when used in a supermarket, the program can output the words written on the labels on products.

The program uses Artificial Intelligence (AI).

(a) Explain how AI is used in the computer program described.

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..... [4]

(b) State **two** social benefits of the use of AI in the computer program described.

1 .....

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2 .....

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[2]

(c) The computer program is released under a commercial software licence.

(i) Describe the features of a commercial software licence.

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..... [3]

(ii) Explain the reasons why an open source software licence might **not** be appropriate for the computer program described.

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..... [3]

5 A programmer is writing a program in a high-level language.

(a) Describe the possible consequences of the programmer **not** joining a professional ethical body.

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..... [3]

6 **(a)** State **two** benefits to a programmer of distributing a program using a shareware licence.

1 .....

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2 .....

..... [2]

(b) Explain why it is important for a programmer to join a professional ethical body.

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..... [4]

7 A computer programmer is learning about the importance of acting ethically at work.

(a) Explain the reasons why the computer programmer needs to act ethically towards colleagues and the public.

Colleagues .....

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The public .....

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[4]

(b) The programmer writes a program to be used in a business. The source code of the program can be modified by the business. The business pays the programmer for the maintenance and security updates.

(i) Identify **one** suitable type of software licence for the program **and** describe the reasons for your choice.

Type of software licence .....

Reasons for choice .....

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[3]

(ii) Explain the reasons why the programmer should copyright their program.

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..... [2]

8 A computer programmer writes programs that are distributed with a software licence.

(a) Complete the table by writing the type of software licence each statement describes.

Statement	Software licence
A fee is charged for the software. The source code cannot be accessed. Users do **not** get a free trial.	....
Users can try the software before buying it. Users may **not** be able to access all features during the trial.	....
Software is usually free of charge. Users can access the source code and alter the program to their needs.	....

[3]

(b) Explain the reasons why it is important for the computer programmer to join a professional ethical body.

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..... [3]

9 **(a)** Complete the table by describing each term.

Term	Description
Copyright	....
Open Source (Initiative)	....
Shareware	....
Software Licence	....

[4]

(b) Describe the purpose of a code of conduct.

.... [2]

10 A programmer is developing a computer program.

(a) The programmer releases the program as Free Software.

Describe what is meant by Free Software.

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..... [2]

(b) A user downloads the computer program from the internet.

State what should be included as part of the download to make sure the program is authentic.

..... [1]

11 A digital video camera records students in a classroom. The data is transferred from the digital video camera over the internet to a server. Artificial Intelligence (AI) is used to analyse the video on the server to identify when students are interacting with the lesson and the teacher.

(a) Describe **one** ethical impact of this use of AI in the classroom.

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..... [2]

12 Compilers and interpreters are used to translate programs written in a high-level language into a low-level language.

(a) A programmer is developing an Operating System (OS).

Identify **two** types of software licence the programmer can use to allow other people to edit and redistribute the OS.

1 .....

2 .....

[2]

END OF QUESTION PAPER